Questions: Introduction to set theory

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Summary

A selection of questions for the study guide on introduction to set theory.

Before attempting these questions, it is highly recommended that you read Guide: Introduction to set theory.

Q1

Decide whether the following sets are finite, countably infinite or uncountable.

1.1. $\{1, 2, 3, 4, 7, 8, 11\}$

1.2. $\mathbb{N}$

1.3. $\{x \in \mathbb{Q} : \frac{1}{9} \leq x \leq \frac{8}{15}\}$

1.4. $\mathbb{R}$

1.5. $(-1.5, 6)$

1.6. $\{x \in \mathbb{Z} : x \geq 8\}$

Q2

Are the following statements true or false?

2.1. $\{1, 3, 9\} \subseteq \{1, 2, 3, 5, 8, 9\}$

2.2. $\{2, 7, 16, 20\} \subseteq \{1, 3, 5, 6, 7, 16, 18, 20\}$

2.3. $\{\frac{1}{3}, 6, 19, \frac{41}{2}\} \subseteq \mathbb{Z}$

2.4. $\{1.6, 1.8, 2.3, 4\} \subseteq [1.6, 5]$

2.5. $\{1.6, 1.8, 2.3, 4\} \subseteq (1.6, 5]$

2.6. $\mathbb{Z} \subseteq \mathbb{R}$.

2.7. $\{2, 4, 8, 15\} \subseteq \{x : \frac{x}{2} \in \mathbb{Z}\}$

2.8. $\{6, 18, 24, 27\} \subseteq \{x : \frac{x}{3} \in \mathbb{Z}\}$

Q3

Write down the power sets of the following:

3.1. $A = \{1, 2, 3\}$.

3.2. $B = \{\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\}$.

3.3. $C = \{1, 3, 6, 9\}$

3.4. $D = \{\emptyset, 4\}$.

3.5. $E = \{4.5, 8, 9, 10\}$

3.6. $F = \{\sqrt{2}, 5, \frac{16}{3}, 18.2\}$

Q4

For each power set in **Q3**, write down its cardinality.

After attempting the questions above, please click this link to find the answers.

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